

3. Autoencoders

Q.1 Explain the role of encoder in an autoencoder.

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- Autoencoder is a type of artificial neural networks that learns to represent data in a compressed form and then reconstruct it as closely as possible to the original input.
- Autoencoder follow an encoder-decoder architecture.
- The encoder is the part of the network that takes the input data and compresses it into a smaller lower-dimensional representation.
- Input layer: This is where the original data enters the network.
- Hidden layer: These layers apply transformation to the input layer data. The encoder's goal is to extract essential features and reduce data's dimensionality.
- Output of Encoder: The encoder outputs a compressed version of the data often called the latent representation or encoding.

Q.2 Describe different types of autoencoders.

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- Autoencoders comes in several types each suited for different tasks and with unique features.

- Types:

- 1) Undercomplete autoencoder
- 2) Overcomplete autoencoder
- 3) Denoising autoencoder
- 4) Variational autoencoder.

1) Undercomplete Autoencoder

⇒ An autoencoder whose code dimension is less than the input dimension is called undercomplete.

- It aims to learn from compressed representation.

- It minimizes the loss between reconstructed input and original input and also aims to extract features of data in low dimensions.

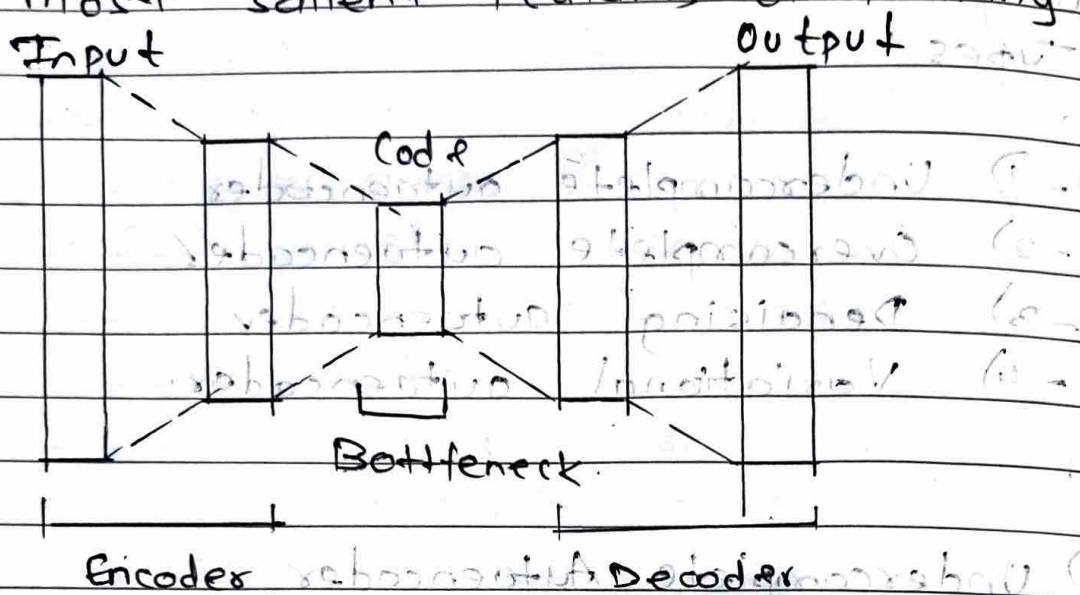
- The learning process is described simply as minimizing a loss function:

- When the decoder is linear and L is the mean squared error, an undercomplete autoencoder learns to span the same subspace as PCA.

- Autoencoders with nonlinear encoder functions F and nonlinear decoder function g can thus

learn a more powerful non-linear generalization of PCA.

- Learning an undercomplete representation forces the autoencoder to capture the most salient features of training data.



2) Overcomplete Autoencoder

⇒ An overcomplete autoencoder has a code dimension greater than the input dimension.

→ This allows network to learn redundant representation of data.

→ The encoder maps the input to the higher-dimensional space.

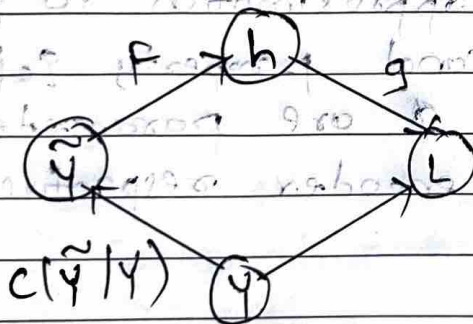
→ The decoder reconstructs the original input.

→ If not regularized properly, the network can learn to copy the input instead of extracting useful features.

B) Denoising Autoencoder (DAE)

F) The DAE is an autoencoder that receives a corrupted data as input and is trained to predict the original uncorrupted data as its output.

- This way the autoencoder can't simply copy the input to its output because the input also contains random noise.
- The noise is subtracted and underlying meaningful data is produced.
- This is called a DAE.
- The DAE training procedure is illustrated as follows.



- 1) Initialize encoder and decoder with random weights.
- 2) Noise is intentionally added to the input data.
- 3) Feedforward the input data through encoder and decoder to get the reconstructed image.
- 4) Calculate the reconstruction loss as defined in our objective function w.
- 5) Do backpropagation and update weights. The goal of training during training is to minimize the reconstruction loss.

4) Variational Autoencoder (VAE)

⇒ VAE makes assumption about the probability distribution of the data and tries to learn a better approximation of it.

→ It uses stochastic gradient descent to optimize and learn the distribution of latent variables.

→ VAEs are often used for generating new data such as creating realistic images to text.

→ It assumes that data is generated by a directed graphical model and tries to learn an approximation to $q_\phi(z|x)$ to the conditional property $q_\theta(z|x)$ where θ and ϕ are parameters of decoder and encoder respectively.

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Q.3 Discuss regularization in autoencoders.

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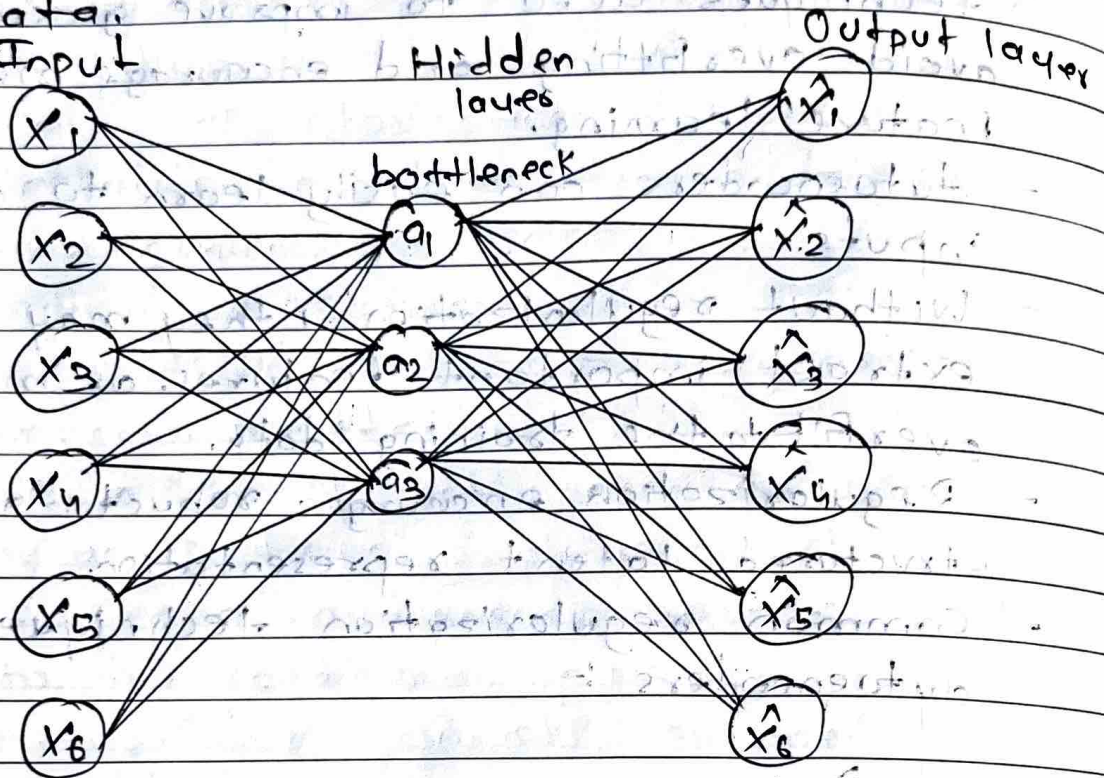
- Regularization in autoencoders refers to techniques used to improve generalization, avoid overfitting and encourage meaningful feature learning.
- Autoencoders can easily learn to copy inputs.
- Without regularization, they may fail to extract important features, or may overfit to the training data.
- Regularization encourages robust, sparse or structured latent representation.
- Common regularization techniques in autoencoders :-
 - 1) Sparse autoencoder
 - 2) Contractive autoencoders
 - 3) Denoising autoencoders.

1) Sparse Autoencoder

⇒ Sparse autoencoders are specific form of autoencoder that's been trained for feature learning and dimensionality reduction.

- As opposed to regular autoencoders, which are trained to reconstruct the input data in the output, sparse autoencoders add a sparsity penalty that encourages the hidden layer to only use a limited no. of neurons at any given time.

- The sparsity penalty causes the model to concentrate on the extraction of the most relevant features from the input data.



- Objective function:

$$L = \|x - \hat{x}\| + \lambda \cdot \text{penalty}(s)$$

2) Contractive Autoencoder

⇒ Another strategy for regularizing an autoencoder is to use a penalty Ω as in sparse autoencoders,

$$L(y, g(F(y))) + \Omega(h(y))$$

but with a different form of Ω :

- This forces the model to learn a function that does not change much when y changes slightly.

- Because this penalty is applied only at training examples, it forces the autoencoder to learn features that capture information about the training distribution.
- An autoencoder regularized in this way is called a contractive autoencoder (CAE).
- This approach has theoretical connections to denoising autoencoders, manifold learning and probabilistic modeling.

Q.4 Applications of Autoencoder:

⇒

1) Image and Audio Compression

⇒ Autoencoders can compress huge images or audio files while maintaining most of the vital information.

- An autoencoder is trained to recover the original picture or audio file from a compressed representation.

2) Anomaly detection

⇒ One can detect anomalies or outliers in dataset using autoencoders.

- Training on autoencoders on a dataset of normal data and any input that the autoencoder cannot accurately reconstruct is called an anomaly.

3) Dimensionality Reduction

⇒ Autoencoders can lower the dimensionality of high-dimensional datasets.

- We can accomplish this by teaching an autoencoder a lower-dimensional data representation that captures the most relevant features.

4) Data Generation

⇒ One can employ autoencoders to

- Employ autoencoders to generate new data similar to the training data.

- One can accomplish this by sampling from the autoencoder's compressed

representation and utilizing the decoder to create new data.

5) Denoising

⇒ One can utilize autoencoders to reduce noise from data.

- We can accomplish this by teaching an autoencoder to recover the original data from a noisy version.

6) Recommender System

⇒ Using autoencoders, we can use users' preferences to generate personalized suggestions.

- We can accomplish this by training an autoencoder to learn a compressed representation of the user's history of system interactions and then utilizing this representation to forecast the user's preference for new items.

Q.5 Explain the Contractive autoencoders, give its applications.

⇒

- Another strategy for regularizing an autoencoder is to use a penalty Ω as in sparse autoencoders,

$$L(y, g(F(y))) + \Omega(h, y)$$

but with a different form of Ω .

- This model forces the model to learn a function that does not change much when y changes slightly.

- Because this penalty is applied only at training examples, it forces the autoencoder to learn features that capture information about the training distribution.

An autoencoder regularized in this way is called a contractive autoencoder (CAE).

- This approach has theoretical connections to denoising autoencoders, manifold learning and probabilistic modeling.
- Working!

- A contractive autoencoder consists of two main components: an encoder and a decoder.

- The encoder compresses the input into a lower-dimensional representation, and the decoder reconstructs the input from this representation.

- The goal is for the reconstructed output to be as close as possible to the original input.
- The training process involves minimizing a loss function that has two terms.
- The first term is the reconstruction loss, which measures the differences between the original input and the reconstructed output.
- The second term is the regularization term, which measures the sensitivity of the encoded representation to the input.
- By penalizing the sensitivity, the CAE learns to produce encodings that do not change much when the input is perturbed slightly, leading to more robust features.
- Applications:

1) Feature learning

⇒ CAEs can learn to capture the most salient features in the data, which can then be used for various downstream tasks such as classification or clustering.

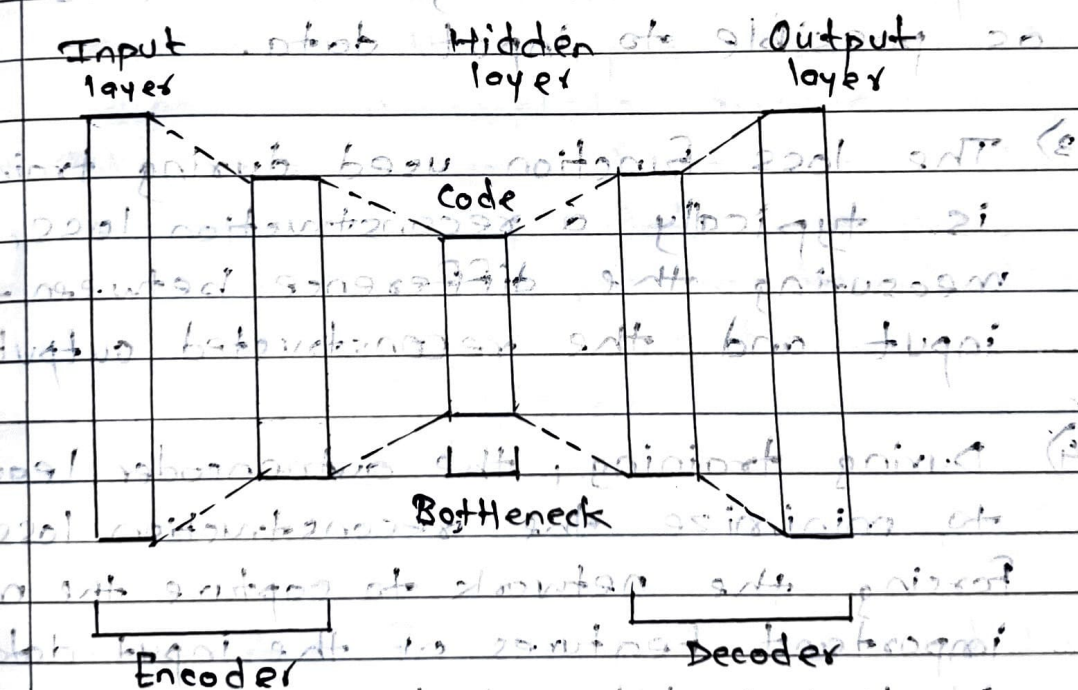
2) Dimensionality Reduction

3) Denoising

4) Data Generation

Discuss Autoencoders with appropriate architecture diagram.

- Autoencoders are a specialized class of algorithms that can learn efficient representation of input data with no need for labels.
- They follow an encoder-decoder architecture.
- The encoder maps the input to latent space and decoder reconstructs the input.
- The architecture of an autoencoder is presented in following figure.



1) Encoder

- ⇒ Input layer takes raw input data.
- The hidden layers progressively reduce the dimensionality of the input, capturing the important features and patterns.
- These layers compose the encoder.

- The bottleneck layer is the final hidden layer where the dimensionality is significantly reduced.

2) Decoder

- ⇒ The bottleneck layer takes the encoded representation and expands it back to the dimensionality of original input.
- The hidden layers progressively increase the dimensionality and aim to reconstruct the original input.
- The output layer produces the reconstructed output, which ideally should be as close as possible to input data.

3) The loss function used during training is typically a reconstruction loss, measuring the difference between the input and the reconstructed output.

4) During training, the autoencoder learns to minimize the reconstruction loss, forcing the network to capture the most important features of the input data in the bottleneck layer.

- After the training process, only the encoder part of autoencoder is retrained to encode a similar type of data used in training process.